Neutron shielding study of a solid state heat transfer method for thermophotovoltaic nuclear reactors

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I. Introduction

Between 2019 and 2024, the efficiency of the top Thermo-Photovoltaïc (TPV) cell has risen from 30 to 44%, largely due to the development of near perfect (>97%) out of band photon backcell reflector [1], [3]. Concomitantly, interest in the application of TPV for the production of electricity in nuclear reactors has risen, with recent papers on TPV nuclear reactor designs [2], [4], as well as patents [13], [14].

A remarkable property of TPV is that it does not need the expansion of gases nor the movement of liquids to produce electricity. Due to its inherent modularity, and low pressure operation, TPV could provide a path toward efficient, yet simple and compact integral modular reactors [21], [22], reducing CAPEX for new nuclear power plants. However, whereas some conventional reactors can operate with a gross efficiency of 44% at a core outlet temperature of 560 ℃ [5], the best current TPV cell needs an emitter temperature of 1435 ℃ to achieve the same efficiency [3]. They also have to be highly shielded from ionizing radiation coming from a nuclear reactor [7], [11] (see II.F)

In order to fully realize the potential of nuclear TPV taking into account the challenge of very high temperature and radiation shielding, this paper explores an original method of heat transfer operating in vacuum, relying on a solid state heat transfer method between the fuel and TPV cells situated in a zone sufficiently shielded from ionizing radiations. Thus, heat transfer within the reactor could be achieved without the use of high pressures, corrosive or otherwise reactive gases or liquids, possibly enabling the design of simpler modular integral reactors.

Using Monte Carlo neutron transport simulation, this paper investigates whether such heat transfer method paired with the suitable shielding material can reduce the neutron Displacement Damage Dose (DDD) in the TPV cells to acceptable levels within an economically justifiable shield design, and which geometry / material would be most suitable for that purpose. This paper focuses on the materials and geometries that provide effective neutron shielding, the additional considerations gamma shielding is not covered by the present paper.

II. Methods

A. Description of the heat transfer method

The heat transfer method discussed in this paper works in the following way: 1) The solid Rotary Heat Shuttle (RHS) is a flat disc or a conical shaped surface rotating in vacuum, 2) When they face each other, the fuel elements radiate heat onto the RHS, and the RHS physically moves that heat during its revolution. 3) When the RHS is in front of the Neutron Shield Moderator / Neutron Shield Absorber, a highly reflective surface on the NSM/NSA may be placed to insulate it from radiative heat transfer from the RHS if the NSM material cannot be at the same temperature as the RHS 4) When the RHS passes in front of the TPV cells, it radiates some of its stored heat onto them.

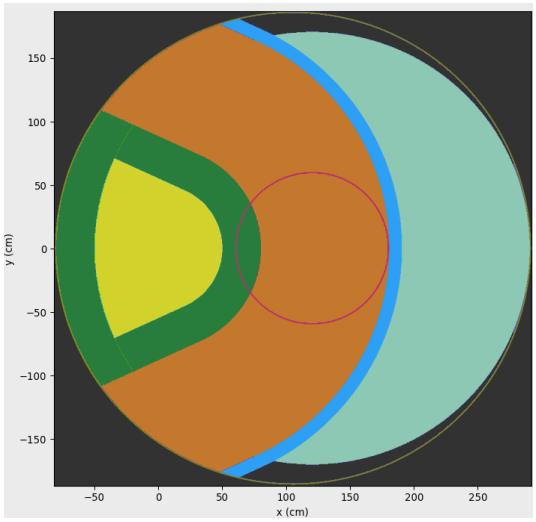
The RHS material is Graphite, chosen for its neutron transparency, low activation, and high temperature resistance, the same qualities that make it the material of choice in HTGR designs operating in the same temperature range. In addition, Graphite can be easily coated with Silicon Carbide (SiC), the emitter chosen for illuminating the reference TPV cells [3] used to parametrize the simulation.

B. Parametric reactor geometry

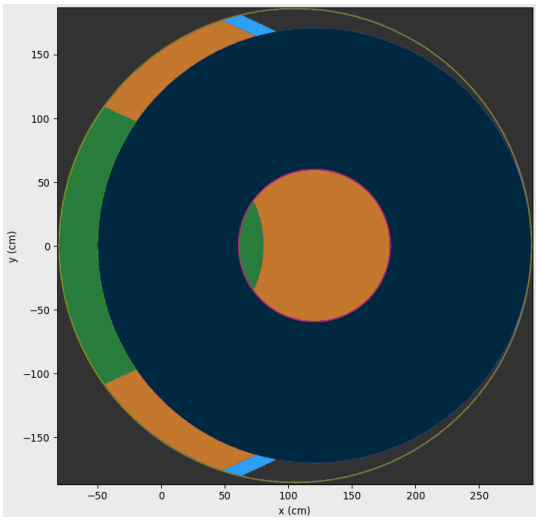
In order to facilitate the setup of OpenMC simulations for calculating DDD on TPV cells, the RIMAG software was created [15]. Using a reduced number of parameters, it accommodates changes of Materials, new shielding layers and dimensioning parameters, including the conical pitch of the RHS to provide for a curved (conical) surface that breaks line of sight from the active zone to the TPV cells. For all simulated configurations in this paper, the simulated reactor was critical with at least a few betas of reactivity margin. Reactor power is deduced by RIMAG from the surface area of TPV cells generated. Average ΔT between the fuel and RHS is calculated based on reactor power and fuel surface area. The source code is available in the accompanying GitHub repository: [https://github.com/creeperdeking/RIMAG/tree/research_paper]

Within the simulation, the reactor “universe” is a z-cylinder, containing a stack made of “fixed” layers

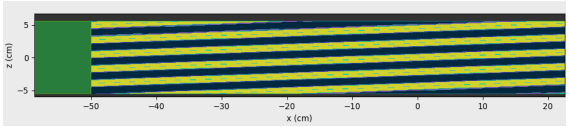
containing fuel, reflector, shielding and TPV cells, alternating with “rotary” layers containing the RHS, forming an infinite vertical stack using a periodic boundary condition on the top and bottom surface of the “universe”, and vacuum boundary condition on the lateral surface.



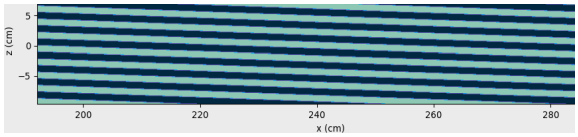
a)



b)



c)



d)

Figure 1. Cross sectional views of reactor geometries generated using RIMAG (a,b) Cross-sectional view of the reactor in the xy plane showing (a) “fixed” layer, (b) “rotary” layer (conical pitch = 0, i.e flat disk geometry). (c,d) Cross-sectional view of the reactor in the xz plane (conical pitch = 2°). Colors denote material regions: [yellow/Nuclear fuel], [green/graphite reflector], [orange/NSM], [blue/NSA], [turquoise/TPV cell cooling sandwiched by thin InGaAs layer], [dark blue/RHS]

The nuclear fuel is a simplified TRISO layer, made of a thin 9.75% enriched UO2 layer sandwiched by graphite, with less than 5% UO2 in the overall TRISO layer by volume. In order to offer a conservative DDD estimation, the space sandwiched between TPV cells is filled with vacuum so that no material would absorb neutrons and possibly bias the simulated DDD downward. However, in a realistic design, water could flow between the solar cells to cool them. This would reduce the mean free path of neutrons within this area of the reactor. In test simulations meant to estimate this effect, replacing the vacuum between solar cells by water resulted in an order of magnitude smaller DDD for a given geometry. The precise design of the TPV cells is important and could decrease by roughly 1/10 the NSM thickness.

C. Choice of reference Thermo-Photovoltaïc cell

While the most efficient operating emitter temperature was experimentally determined to be 44% at 1435 ℃ [3], because of the difficulty of developing and certifying new high temperature nuclear fuel, a more conservative emitter temperature was used in this study based on the temperature attained by the TRISO fuel from the AGR-2 irradiation campaign fuel temperature average [6]. The reference fuel temperature chosen was 1360 ℃. Using this fuel temperature as input, and emitter emissivity of 0.96 [3], the temperature of the RHS was estimated by RIMAG to be between 1178 ℃ and 1265 ℃ for all simulated geometries. Using figure 8b from [1], the highest efficiency TPV cell within this emitter temperature range was identified as having a bandgap between 0.7 and 0.8eV, this is the reason for chosing the 0.74 eV InGaAs cell from [3]. Within this temperature range, it averaged an efficiency of about 33%, lower than its optimum at 1435 ℃ but still within the range typical of modern nuclear power facilities. Taking into account the view factor of 38% for measured power density in [3], the 100% view factor equivalent power density for the cell is 0.89 W/cm2 electrical, out of 2.7 W/cm2 total radiative power absorbed by the cell.

An acceptable useful lifetime of TPV cells within the reactors was arbitrarily defined as >90% of original power density (Pmax) after 10 years of continuous irradiation within the reactor at nominal power.

D. Determining acceptable Displacement Damage Dose threshold for reference cell

Using the fluence to Pmax relationship shown in figure 6d of [7], the total neutron irradiation dose intensity that would result in a 10% reduction of Pmax within the Xi'an Pulsed Neutron Reactor (XPNR) was read as approximately 4.56×10^{10} n/cm2. In order to make the result independent of the specific neutron spectrum of the XPNR, Table 1 of [7] provides the corresponding total Displacement Per Atom (DPA_{XPNR}) of 2.890×10^{-11} for the aforementioned acceptable total irradiation dose density. Now, to convert DPA_{XPNR} into an equivalent Displacement Damage Dose (DDD) for the InGaAs cell material, we use the Norgett, Robinson, and Torrens (NRT) model, according to ASTM guidance [8]:

$$dpa \approx \frac{\beta}{2T_d}\frac{DDD}{N_{atoms/g}} \quad \text{Eq. 1}$$

$$DDD \approx dpa \times N_{atoms/g} \times \frac{2T_d}{\beta} \quad \text{Eq. 2}$$

With β = 0.8 [8] and T_d the displacement threshold energy chosen at a conservative 10 eV, the value defined in the ASTM E722 GaAs neutron damage-function convention [9], which is also used to compute DDD within the OpenMC simulation.

For lattice-matched $In_{0.53}Ga_{0.47}As$, with $M \approx 168.545\ g/mol$, we calculate $N_{atoms/g}$:

$$N_{atoms/g} = \frac{2N_A}{M} \approx 7.146 \times 10^{21}\ atoms/g \quad \text{Eq. 3}$$

DDD threshold for a 10% reduction in Pmax within $In_{0.53}Ga_{0.47}As$ cell is thus calculated as $DDD \approx 5.16 \times 10^{6}$ $MeV \cdot g^{-1}$.

E. Calculating Displacement Damage Dose in OpenMC

In order to determine the effective NSM thickness required to cross the DDD threshold experimentally derived from the XPNR's data in II.D, the OpenMC neutron transport code is used [16]. In order to obtain the simulated DDD value, a flux score tally is applied on the cell representing the TPV cell within the simulation. In OpenMC, flux score is a volume integrated track-length flux with units of particle.cm per source particle [17]. Dividing by volume and multiplying by source strength gives flux in $particles \cdot cm^{-2} \cdot s^{-1}$ [18]. Applying the filter openmc.EnergyFunctionFilter to the tally multiplies that flux score by the tabulated function of incident energy provided to EnergyFunctionFilter [19]. The tabulated function appropriate for the TPV cell material relates incident energy with DDD in $MeV \cdot cm^2 \cdot g^{-1}$ [10]. Data for the tabulated function was obtained using the ASTM E722 GaAs neutron damage-function convention, as implemented in the SR-NIEL neutron NIEL calculator, in which the GaAs displacement threshold energy is E_d = 10 eV [11], [12]. After the simulation is done, the flux tally is fetched, divided by cell volume and multiplied by source strength to give a DDD flux in $MeV \cdot g^{-1} \cdot s^{-1}$. In order to convert it to the same units as the threshold from II.E, this flux is multiplied by irradiation time (10 years) to get a DDD value on the cell in $MeV \cdot g^{-1}$ that can be compared with the acceptable Displacement Damage Dose threshold for reference cell calculated in II.E

F. Neutron shield design

To a first approximation, typical fluence outside a nuclear reactor active zone is on the order of $10^{13} n/cm2/s$ [11]. A neutron flux of this magnitude on the TPV cell would induce a degradation of 50% of PMax within one second [7]. A reduction of 11 orders of magnitude in neutron intensity between the core and the TPV cells might thus be expected to keep Pmax losses below 10% for 10 years. A typical solution to provide neutron shielding of this magnitude where space is a constraint is the use of the so-called multilayer shield, with layers arranged perpendicularly to the radial direction from the source [12]. A first material, with a high moderating power is placed forward, while the second material exhibits a high neutron absorption cross section. Scattering of high energy neutrons is also significant for increasing opacity to neutrons in the first layer. For simplicity, in this paper the shield consists of only one NSM, and one B_4C Neutron Shield Absorber (NSA). Absorber thickness was set constant for all simulations at 10 cm. This paper focuses on neutron shielding, acknowledging that a realistic shield would contain more layers, at least to shield from secondary gammas emitted by the B_4C layer. Six materials are explored offering tradeoffs between moderating power, cost and requirement for cooling at nominal reactor temperature in order to prevent phase change or high pressures. OpenMC neutron transport is done for progressively higher NSM thickness at a fixed conical pitch of 15° for the RHS in order to tabulate the neutron attenuation trend.

TABLE 1: RAW MATERIAL COST FOR TESTED NEUTRON SHIELD MODERATOR MATERIALS

NSM material	USD/kg
Water	1.7×10^{-3} [28]
High Density PolyEthylene (HDPE)	1.2 [29]
Titanium Hydride (TiH2)	9.0×10^{1} [30]
Beryllium Oxide (BeO)	5.0×10^{2} [31]
Lithium Oxide (Li2O)	1.0×10^{2} [35], [36]
Graphite	5.9×10^{-1} [26]

All prices were adjusted to 2025 USD using CPI from [34]

All results are then extrapolated using an exponential fit to deduce the NSM thickness at which sufficient shielding would be provided. The reason for this extrapolation is that simulating all the way to sufficient shielding requires exponentially more computing power in Monte Carlo neutron simulations, in practice the level of attenuation required could not be computed for this work within reasonable error bounds.

In some data series, a discontinuity was observed, with DDD reducing more slowly starting at a threshold thickness. When this was observed the exponential attenuation extrapolation was done using the last three data points to avoid underestimating the total required NSM thickness.

G. *Effect of the conical pitch of the Rotary Heat Shuttle*

Pitch angles for the conical RHS at 0°, 1°, 2°, 5°, 15° and 45° were simulated at increasing NSM thickness, using water as the NSM material. Results were extrapolated to get estimated NSM thickness for the required DDD attenuation for each pitch angle in order to test the hypothesis that a flat RHS (conical pitch = 0) would not allow for practical neutron attenuation.

It is expected that a flat RHS would act like a transparent "slit" from the point of view of a fraction of the neutrons coming from the active zone. They would avoid going through the NSM altogether, causing the neutron attenuation to asymptote to a linear reduction with increasing NSM thickness (linear decrease in angular line of sight with increasing slit depth). On the other hand, fully breaking line of sight using a curved surface was expected to allow the exponential attenuation characteristic of the thick shield approximation [20] to be recovered, thus allowing for effective neutron shielding.

H. *Determining a conservative shield cost threshold*

In order to provide a first conservative screening on the different options for the NSM from a cost standpoint, the RIMAG software was used to calculate the NSM volume in cm3 / kW, which was converted to $/kW using the costs from Table 1. This cost was compared to the average worldwide equipment only cost of the nuclear island in $/kW in order to screen NSM options that would exceed this conservative cost threshold. The nuclear island cost was estimated using the average capacity-weighted overnight cost for the entire plant [27] giving a cost in 2025 dollars of 4,594 $/kW . The equipment only fraction of that cost was estimated from [32] as 21% of overnight cost, while the reactor plant equipment share of overnight cost was estimated to be 45% from [33]. Thus, the equipment-only overnight cost fraction of the nuclear island is estimated to be 9.5% of overnight cost, yielding a value of 440 $/kW which is the threshold chosen for screening candidate NSM materials.

III. RESULTS

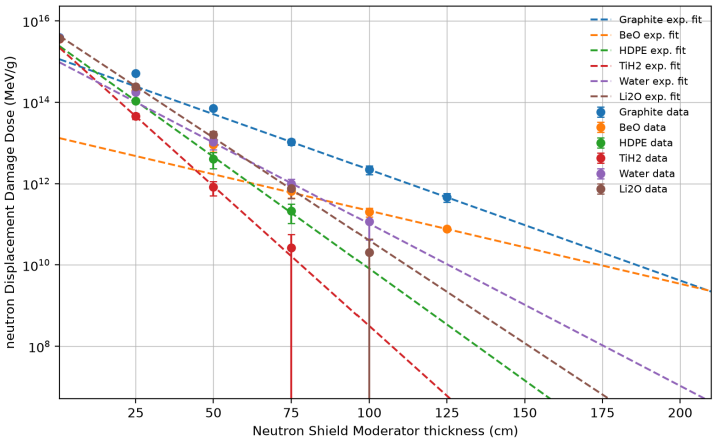


Figure 2. Plot of DDD against NSM shield thickness for candidate NSM materials for geometries at constant 15 ° Conical pitch

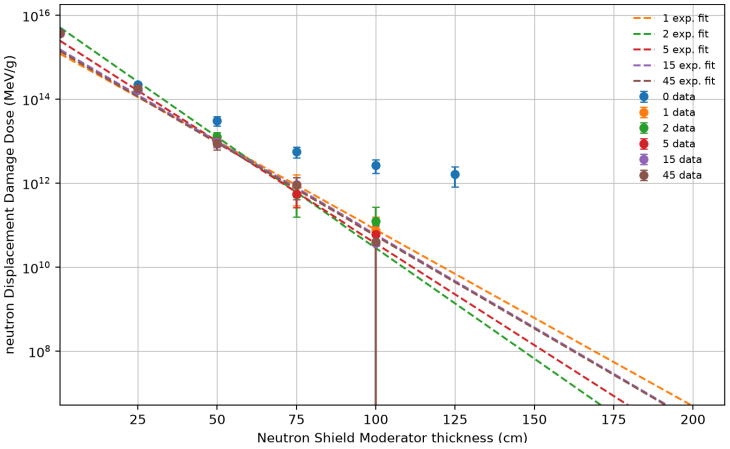


Figure 3. Plot of DDD against NSM shield thickness for various conical pitch angle values with water NSM material.

TABLE 2: SIMULATED REQUIRED NSM THICKNESS FOR EACH MATERIAL AND CALCULATED MATERIAL COST IN $/KW

NSM material	Extrapolated total thickness (cm)	$/kW
Water	208	5.5×10^{-3}
High Density PolyEthylene (HDPE)	158	2.7
Titanium Hydride (TiH2)	126	6.1×10^{2}
Beryllium Oxide (BeO)	357	9.2×10^{3}
Lithium Oxide (Li2O)	177	5.1×10^{2}
Graphite	307	5.2

IV. DISCUSSION

From Figure 2, The log-linear derivative of the attenuation power gives extrapolated NSM thicknesses for the required DDD threshold between 1.2 and 3.6 meters. In Table 2, cost data were deduced from the required NSM thicknesses obtained for creating Figure 2 and the costs from Table 1. When comparing these costs to the chosen threshold in section II.H, TiH2, BeO and Li2O exceeded the threshold, making them unlikely candidates for a competitive reactor design. On the other hand, Water, HDPE and Graphite were all more than two orders of magnitude below the conservative cost threshold. Graphite is particularly attractive due to its ability to perform well structurally at the reactor nominal temperature without requiring complex active cooling and very compact reflective insulation, which would add cost and complexity to designs using water, HDPE or TiH2 (because of dehydrogenation well below the temperature of the RHS). Where reducing reactor size is of the highest priority, water and HDPE would be preferred, although cooling and insulation systems would reduce the effective density of HDPE in practice. Graphite is a very good neutron reflector, making it the NSM would provide advantages in terms of neutron economy.

In Figure 3, for the 0° pitch data points, the hypothesis that a flat RHS would significantly depart from exponential attenuation is borne out. Curved (conical) geometries do recover exponential attenuation for all tested non zero conical pitch, with no clear pattern between different pitch values discernable with the currently simulated datapoints.

V. CONCLUSION

Given the conical RHS design considered, among cost competitive options, hydrogenated shield moderators, in particular Water and HDPE were found to be the NSM materials that allow for the most compact shields. However they would need complex active cooling and very compact reflective insulation, while Graphite would be both cost competitive and less complex, at the expense of a larger shield, and thus bigger reactor.

Making the RHS flat (pitch=0°) prevents the neutron DDD on the cells from decreasing exponentially with shielding thickness, making a flat geometry unsuitable for shielding. This is coherent with a theoretical model assuming some neutrons are able to move in a straight line from the fuel to the solar cells using the RHS as a neutronically “transparent” slit between the neutron shield. On the other hand, exponential attenuation can be recovered provided the RHS is conical, for all measured non zero pitch values.

In conclusion, efficient shielding can be provided cost effectively within a distance on the order of 1.5 to 3 meters. This opens the way for further investigations of this original concept for realizing practical TPV nuclear reactors, first by confirming and refining this paper’s results by computing additional data points, studying gamma radiation shielding, as well as an optimisation of neutronic, thermodynamic, and mechanical constraints a practical commercial reactor would be subject to.

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Code availability. The parametric OpenMC reactor geometry generator used to create the reactor models in this study is publicly available at[https://github.com/creeperdeking/RIMAG/tree/research_paper] The repository includes instructions for regenerating the OpenMC geometry files used in the study.